

Challenge (Habanero)

- Q1.** What is the overlap region for $x + y \geq 4$ and $2x - y \leq 3$?
- (A) Triangle
(B) Quadrilateral
(C) Pentagon
(D) Parallelogram
(E) Hexagon
- Q2.** A gamer's score s and time t are related by $s \geq 2t - 5$ and $s \leq 3t + 2$. What is the feasible region?
- (A) Below the line $s = 2t - 5$
(B) Between $s = 2t - 5$ and $s = 3t + 2$
(C) Above the line $s = 3t + 2$
(D) Left of the line $t = 2$
(E) Right of the line $t = 5$
- Q3.** For the system $x - 2y < 3$ and $x + 2y > 5$, which quadrant has no solution?
- (A) I
(B) II
(C) III
(D) IV
(E) All quadrants have solutions
- Q4.** A coding tournament has two events: x hours of individual coding and y hours of team coding, with $x + y \leq 8$ and $x \geq 2$. What is the lower bound for y ?
- (A) 0 hours
(B) 2 hours
(C) 4 hours
(D) 6 hours
(E) 8 hours
- Q5.** In the game *GalacticConquest*, the number of spaceships s and the budget b are related by $s \leq 2b - 10$ and $s \geq b - 5$. What is the least budget for 10 spaceships?
- (A) 10
(A) 12
(A) 15
(A) 20
(A) 25
- Q6.** The region defined by $x + y \geq 2$ and $x - 2y \leq 1$ is a
- (A) Triangle
(B) Quadrilateral
(C) Parallelogram
(D) Rectangle
(E) Hexagon
- Q7.** A data analyst's time t and data points d are related by $t \geq d - 3$ and $t \leq 2d + 1$. What is the overlap region?
- (A) $d \geq 2$ and $t \geq d - 3$
(B) $d \leq 2$ and $t \leq 2d + 1$
(C) $t \geq d - 3$ and $t \leq 2d + 1$
(D) $d \geq 1$ and $t \geq 2$
(E) $d \leq 3$ and $t \leq 5$
- Q8.** A gaming console's processing power p and memory m are related by $p \leq 3m - 2$ and $p \geq 2m + 1$. What is the constraint on m ?
- (A) $m \geq 1$
(B) $m \leq 2$
(C) $m \geq 3$
(D) $m \leq 4$
(E) $m \geq 5$

Open Ended Questions (FRQ)

- Q9.** Graph the system $x + 2y \geq 4$ and $x - y \leq 2$, and identify the feasible region. Show all work.
- Q10.** A programmer's coding time t and lines of code l are related by $t \geq 2l - 10$ and $t \leq 3l + 5$. Find the feasible region and interpret the results in the context of coding. Show all work.

Chef's Tips

- Tip 1: Ensure students understand the concept of overlapping regions
- Tip 2: Emphasize the importance of interpreting results in context
- Tip 3: Encourage students to show all work and justify their answers